EASY · ALGEBRA · ANSWER KEY

Algebra

16

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ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

B

B. $3, 4$

Choice B is correct. If a point x, y lies on both lines in the graph of a system of two linear equations, the ordered pair x, y is a solution to the system. The graph shown is the graph of a system of two linear equations, where the two lines in the graph intersect at the point $3, 4$. Therefore, the point $3, 4$ lies on both lines, so the ordered pair $3, 4$ is the solution to the system.

Choice A is incorrect. The point $2, 3$ lies on one, not both, of the lines in the graph shown.

Choice C is incorrect. The point $4, 5$ lies on one, not both, of the lines in the graph shown.

Choice D is incorrect. The point $5, 6$ lies on one, not both, of the lines in the graph shown.

Q2

A

A. The interior designer earned $\$ 68$ per hour consulting last week.

Choice A is correct. It's given that $68x + 85y = 1{,}258$ represents the situation where an interior designer earned a total of $\$ 1{,}258$ last week from consulting for x hours and drawing up plans for y hours. Thus, $68x$ represents the amount earned, in dollars, from consulting for x hours, and $85y$ represents the amount earned, in dollars, from drawing up plans for y hours. Since $68x$ represents the amount earned, in dollars, from consulting for x hours, it follows that the interior designer earned $\$ 68$ per hour consulting last week.

Choice B is incorrect. The interior designer worked y hours, not 68 hours, drawing up plans last week.

Choice C is incorrect. The interior designer earned $\$ 85$ per hour, not $\$ 68$ per hour, drawing up plans last week.

Choice D is incorrect. The interior designer worked x hours, not 68 hours, consulting last week.

Q3

0.2

Accept also: 1/5

The correct answer is $.2$. Subtracting 2.6 from each side of the given equation yields $x = 0.2$. Therefore, the value of x that's the solution to the given equation is 0.2. Note that .2 and 1/5 are examples of ways to enter a correct answer.

Q4

B

B. 12

Choice B is correct. For the graph shown, the x-axis represents temperature, in kelvins, and the y-axis represents the estimated pressure, in $\text{pounds per square inch psi}$. The estimated pressure of the argon when the temperature is 600 kelvins can be found by locating the point on the graph where the value of x is equal to 600. The graph passes through the point $600, 12$. This means that when the temperature is 600 kelvins, the estimated pressure is 12 psi.

Choice A is incorrect. This is the estimated pressure, in psi, of the argon when the temperature is 300 kelvins, not 600 kelvins.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the temperature, in kelvins, of the argon.

Q5

C

C. -4, -3

Choice C is correct. The solution to a system of linear equations is represented by the point that lies on the graph of each equation in the system, or the point where the lines intersect on a graph. On the graph shown, the two lines intersect at the point -4, - 3. Therefore, the solution to the system is -4, - 3.

Choice A is incorrect. This is the y-intercept of the graph of one of the lines shown, not the intersection point of the two lines.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

D

D. 18

Choice D is correct. It's given that the shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 220 pounds. Let x represent the number of 5-pound boxes and y represent the number of 10-pound boxes in the shipment. Therefore, the equation $5x + 10y = 220$ represents this situation. It's given that there are 13 10-pound boxes in the shipment. Substituting 13 for y in the equation $5x + 10y = 220$ yields $5x + 1013 = 220$, or $5x + 130 = 220$. Subtracting 130 from both sides of this equation yields $5x = 90$. Dividing both sides of this equation by 5 yields 18. Thus, there are 18 5-pound boxes in the shipment.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the number of 10-pound boxes in the shipment.

Q7

B

B. 18

Choice B is correct. Subtracting 12 from each side of the given equation yields $3x - 12 = 30 - 12$, or $3x - 12 = 18$. Therefore, the value of $3x - 12$ is 18.

Choice A is incorrect. This is the value of $x - 12$, not $3x - 12$.

Choice C is incorrect. This is the value of $x + 12$, not $3x - 12$.

Choice D is incorrect. This is the value of $3x + 12$, not $3x - 12$.

Q8

50

The correct answer is 50. It's given that the function f gives the total number of people on a company retreat with x managers. It's also given that 7 managers are on the company retreat. Substituting 7 for x in the given function yields $f7 = 77 + 1$, or $f7 = 50$. Therefore, there are a total of 50 people on a company retreat with 7 managers.

Q9

6

The correct answer is 6. The first equation in the given system is $x = 8$. Substituting 8 for x in the second equation in the given system yields $8 + 3y = 26$. Subtracting 8 from both sides of this equation yields $3y = 18$. Dividing both sides of this equation by 3 yields $y = 6$. Therefore, the value of y is 6.

Q10 A

A. $x + 3y = 240$

Choice A is correct. It's given that for a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water. Since the group bought x one-liter bottles of water, the total number of liters bought from x one-liter bottles of water is represented as $1x$, or x. Since the group bought y three-liter bottles of water, the total number of liters bought from y three-liter bottles of water is represented as $3y$. It's given that the group bought a total of 240 liters; thus, the equation $x + 3y = 240$ represents this situation.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the group bought x three-liter bottles of water and y one-liter bottles of water, for a total of 240 liters of water.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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